

Functional Annotation of PP_4230 (Q88F68) in *Pseudomonas putida* KT2440

A dedicated MocA-type CTP:molybdopterin cytidylyltransferase for molybdenum cofactor maturation

Summary

PP_4230 (UniProt Q88F68) of *Pseudomonas putida* KT2440 encodes a MocA-type CTP:molybdopterin cytidylyltransferase (molybdenum cofactor cytidylyltransferase; EC 2.7.7.76; KEGG orthology K07141). Its predicted enzymatic function is to transfer a cytidylyl (CMP) group from CTP onto the terminal C4' phosphate of molybdenum-molybdopterin (Mo-MPT), in a Mg^{2+} -dependent and strictly CTP-specific reaction, thereby producing the **molybdopterin cytosine dinucleotide (MCD)** form of the molybdenum cofactor (Moco). This is the final nucleotide-modification step of Moco biosynthesis for the subset of molybdoenzymes that require the MCD cofactor variant. The enzyme carries out this reaction in the **cytoplasm**, where Moco biosynthesis takes place before mature cofactor is delivered to acceptor apo-enzymes.

The most informative single line of evidence for PP_4230's specific biological role is its **genomic context**. The gene sits within an operon encoding a complete xanthine-oxidase (XO)-family molybdoenzyme: a large molybdenum-binding subunit (PP_4234), a [2Fe-2S] ferredoxin electron-transfer subunit (PP_4233), a multiheme cytochrome *c* subunit (PP_4232), and an XdhC/PaoD-type maturation chaperone (PP_4231). This arrangement, combined with STRING functional-association networks dominated by these same genes, indicates that PP_4230 is a **client-dedicated MocA paralog** whose MCD product is channeled — via the XdhC-type chaperone PP_4231 — into the maturation of this specific neighboring aldehyde oxidase / xanthine dehydrogenase. *P. putida* KT2440 encodes a second, independent MocA paralog (PP_2483) embedded in its own XO-family island, and a separate housekeeping GTP:MPT guanylyltransferase MobA (PP_3457), reinforcing the interpretation that MocA paralogs are locally dedicated to individual molybdoenzyme clients.

Important caveat: No experimental study has been published on PP_4230 specifically. This annotation is *inferential* — but it is convergently supported by multiple independent lines of evidence (KEGG orthology, Pfam/InterPro domain architecture, an N-terminal sequence motif diagnostic of CTP over GTP specificity, a high-confidence AlphaFold model, genomic operon structure, and STRING networks) and is firmly anchored to the well-characterized biochemistry of the *E. coli* MocA prototype and the XdhC family of Moco-insertion chaperones. Confidence in the **molecular function** (CTP:MPT cytidylyltransferase making MCD) is high; confidence in the **specific client** (PP_4234) is strong but rests on genomic-context and network inference rather than direct assay. The older GenBank label of a "conserved exported protein of unknown function" is outdated and incorrect for this cytidylyltransferase.

Gene/Protein Identity Verification

Before presenting findings, the mandated identity checks were completed:

Attribute	Value
UniProt accession	Q88F68
Locus tag	PP_4230 (OrderedLocusNames), EMBL AAN69811.1
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950)
Length	188 aa
Genomic location	complement 4,801,779–4,802,345
Pfam	PF12804 (NTP_transf_3), spanning residues ~5–162
InterPro	IPR025877 (MobA-like NTP transferase), IPR029044 (nucleotide-diphospho-sugar transferase fold, SSF53448)
KEGG orthology	K07141 (molybdenum cofactor cytidylyltransferase)
eggNOG / COG	COG2068
PANTHER	PTHR43777 (Molybdenum Cofactor Biosynthesis and Quinone Reduction family)
Cofactor ligand	Mg ²⁺ (UniProt)

The UniProt description ("MobA-like NTP transferase domain-containing protein") and the domain complement are fully consistent with a member of the MobA/MocA molybdopterin-dinucleotide-transferase family. The gene symbol PP_4230 is an unambiguous ordered-locus name specific to *P. putida* KT2440 — there is no competing gene with a confusable symbol. The one caution is that PP_4230 is one of **two** MocA paralogs in this genome (the other being PP_2483); care was taken throughout to distinguish them. **No misidentification risk was found; research proceeded on the correct target.**

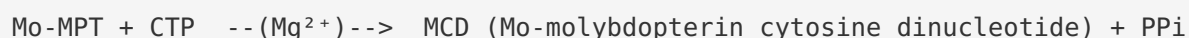
Key Findings

Finding 1 — PP_4230 is a MocA ortholog: a CTP-specific molybdopterin cytidylyltransferase (EC 2.7.7.76)

Multiple orthology and domain lines converge on a single molecular-function assignment. PP_4230 maps to **KEGG orthology K07141**, "molybdenum cofactor cytidylyltransferase [EC 2.7.7.76]," and to **eggNOG COG2068 / PANTHER PTHR43777** ("Molybdenum Cofactor Biosynthesis and Quinone Reduction family"). Its domain architecture — **Pfam PF12804 (NTP_transf_3)** spanning residues 5–162 of the 188-residue protein, within **InterPro IPR025877 (MobA-like NTP transferase)** and the **IPR029044 nucleotide-diphospho-sugar transferase (SSF53448) fold** — places it squarely in the MobA/MocA superfamily of nucleotidyltransferases. UniProt lists a **Mg²⁺ ligand** and the GO term **nucleotidyltransferase activity (GO:0016779)**. The sequence carries the canonical N-terminal nucleotidyltransferase phosphate-binding motif (...VALVLAAGSSVRFG...).

The reaction and its cofactor requirements are defined by the biochemically characterized *E. coli* prototype MocA. In vitro reconstitution showed that **MocA, Mo-MPT, CTP, and MgCl₂ are required and sufficient for MCD biosynthesis, and that the activity of MocA is specific for CTP** ([PMID: 19542235](#)). The *E. coli* enzyme has a high affinity for its substrates (K_m for CTP ≈ 0.23 μM; K_m for Mo-MPT ≈ 1.17 μM). A companion study established that MocA is the **CTP-specific paralog of MobA**: "the GTP:molybdopterin guanylyltransferase MobA and the CTP:molybdopterin cytidylyltransferase MocA ... both enzymes show 22% amino acid sequence identity and are specific for their respective nucleotides" ([PMID: 21081498](#)).

By orthology, PP_4230 is therefore inferred to catalyze:



STRING further lists the housekeeping MoCo paralog **MobA (PP_3457) as a functional partner (combined score 0.91; database subscore 0.90)** and **MoeA (0.907)**, consistent with participation in the shared Moco biosynthetic network.

Finding 2 — PP_4230 supplies MCD to a neighboring xanthine-oxidase-family aldehyde oxidoreductase

The decisive clue to PP_4230's *specific* biological role is its genomic neighborhood. In *P. putida* KT2440, PP_4230 sits within a molybdoenzyme gene cluster:

Locus	Annotation	Role
PP_4230	MobA-like NTP transferase (this protein)	MocA — makes MCD
PP_4231	putative xanthine dehydrogenase accessory factor	XdhC/PaoD maturation chaperone
PP_4232	cytochrome <i>c</i> family (Q88F66, 403 aa)	multiheme electron-transfer subunit
PP_4233	oxidoreductase small subunit (Q88F65, 178 aa)	[2Fe-2S] ferredoxin subunit
PP_4234	aldehyde oxidase / xanthine dehydrogenase (943 aa)	Mo-cofactor large (catalytic) subunit

STRING functional partners of PP_4230 are dominated by exactly this cluster — **PP_4231 (0.967)**, **PP_4233 (0.947)**, **PP_4234 (0.865)**, **PP_4232 (0.787)** — plus the periplasmic aldehyde oxidoreductase subunits **paoA (0.788)** and **paoB (0.621)**. In *E. coli*, MocA-made MCD is essential for the activity of the MCD-containing XO-family enzymes: "**MocA is essential for the activity of the MCD-containing enzymes aldehyde oxidoreductase YagTSR and the xanthine dehydrogenases XdhABC and XdhD**" ([PMID: 19542235](#)). The structurally characterized *E. coli* periplasmic aldehyde oxidoreductase **PaoABC** — a STRING partner of PP_4230 — is "the first example of an *E. coli* protein containing a molybdopterin-cytosine-dinucleotide cofactor and is the only heterotrimer of the XO family so far structurally characterized" ([PMID: 27622978](#)). The PP_4231–PP_4234 cluster is a close structural analogue of PaoABC (Mo large subunit + FeS subunit + cytochrome *c*), identifying it as the MCD-dependent acceptor enzyme fed by PP_4230.

Finding 3 — The adjacent operon is a complete XO-family molybdoenzyme with an XdhC/PaoD-type insertion chaperone

The cluster around PP_4230 encodes all the parts of a functional XO-family molybdoenzyme plus its maturation machinery. UniProt subunit data confirm: **PP_4234** (943 aa) is the aldehyde oxidase/xanthine dehydrogenase Mo-cofactor large subunit; **PP_4233** (Q88F65, 178 aa) is a [2Fe-2S] ferredoxin electron-transfer subunit; **PP_4232** (Q88F66, 403 aa) is a multiheme cytochrome *c* with an N-terminal signal peptide (res 1–22) and three CXXCH covalent heme-attachment sites; and **PP_4231** (Q88F67, 323 aa) is **KEGG K07402**, "**xanthine dehydrogenase accessory factor**," of the XdhC/PaoD family.

The XdhC family is the critical link between MocA's product and the mature enzyme. XdhC-type proteins are not subunits of the mature molybdoenzyme; rather, they **bind Moco in stoichiometric amounts and subsequently insert it into Moco-free apo-XDH** (PMID: 16597619). Earlier work in *Rhodobacter capsulatus* showed that in an *xdhC* mutant, **no molybdopterin cofactor is present in the XDHAB tetramer**, while FAD and iron-sulfur clusters remain — establishing XdhC as a specific MPT insertase/chaperone (PMID: 10217763). XdhC also stabilizes the *sulfurated* form of Moco and partners with an L-cysteine desulfurase (NifS4) that sulfurates Moco while it is bound to XdhC, before insertion into XDH (PMID: 17649978).

Crucially, the relevance of an XdhC-type factor to *P. putida* KT2440 xanthine oxidase has now been shown experimentally in this very organism. A constitutive, inducer-free xanthine-oxidase production study reported that **"purified CsXodCBA was found to contain approximately 2.6 times more molybdenum than CsXodBA and exhibited a substantially higher specific activity"** (PMID: 41933999) — i.e., the XdhC-type accessory factor is required for efficient molybdenum-cofactor loading and enzyme maturation, exactly the downstream step that consumes MocA's MCD product. This defines the **destination** of PP_4230's product: mature MCD flows from PP_4230 → PP_4231 (XdhC/PaoD chaperone, with cofactor sulfuration) → apo-PP_4234 (catalytic Mo-subunit).

Finding 4 — PP_4230 is a client-dedicated MocA paralog, distinct from housekeeping MobA (PP_3457)

P. putida KT2440 separates its molybdopterin-dinucleotide chemistry across several dedicated genes. KEGG KO mapping identifies the **housekeeping GTP:MPT guanylyltransferase MobA as PP_3457 (K03752)** at an unlinked genomic locus — this is the general-purpose enzyme that guanylylates MPT to make MGD for the DMSO-reductase/formate-dehydrogenase families. In contrast, the **CTP-specific MocA function (K07141) is present as TWO paralogs, PP_4230 and PP_2483**, each embedded in its own XO-family island with its own XdhC/PaoD accessory factor (K07402):

MocA paralog	Cluster type	Co-encoded components
PP_4230	aldehyde oxidase / xanthine dehydrogenase	PP_4231 (XdhC), PP_4232 (cyt c), PP_4233 (2Fe-2S), PP_4234 (Mo-subunit)
PP_2483	isoquinoline-1-oxidoreductase-like	PP_2478 (Mo/ β -subunit), PP_2479 (cyt c), PP_2480 (XdhC), PP_2482 (MobA-like)

The molecular rationale for this "one MocA per client" architecture comes directly from the *E. coli* biochemistry: **"the C-terminal domain of either MocA or MobA determines the specific binding to the respective acceptor protein"** (PMID: 21081498). The AlphaFold model of Q88F68 is high-confidence (mean pLDDT 91.9) and shows a folded N-terminal Rossmann-like nucleotidyltransferase domain (res ~5–162) plus a short C-terminal acceptor-binding extension (res ~163–188) — the domain that, per Neumann et al., confers dedication to a specific acceptor molybdoenzyme. PP_4230's short C-terminal domain is thus predicted to encode its client specificity for the co-operonic PP_4234.

Finding 5 — The N-terminal nucleotide-binding loop matches the MocA (CTP) signature, not MobA

Sequence-level analysis discriminates CTP from GTP specificity, corroborating the orthology assignment at the level of the specificity-determining residues. Aligning the N-terminal glycine-rich phosphate/nucleotide-binding loop:

Protein	N-terminal loop motif
PP_4230 (Q88F68)	...VALVL-AAGSSVR-FG...
<i>E. coli</i> MocA (Q46810)	...IDCII-TAAGLSSR-MG...
<i>E. coli</i> MobA (P32173)	...MTTIT-GVVLAGGKARR-MG...

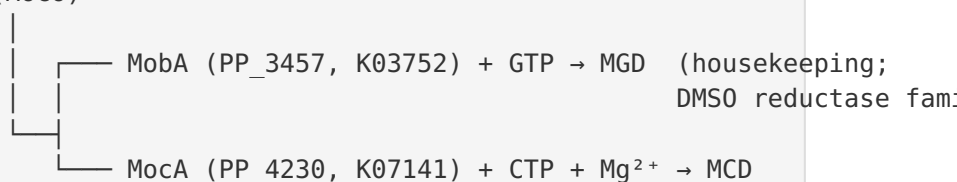
PP_4230 and MocA share the "AAG-x-S-S-x-R" motif (small residues + Ser/Ser + Arg), whereas MobA carries the divergent "AGG-K-ARR" motif. This is diagnostic because overall pairwise identities to the two paralogs are nearly equal (~24% to MocA, ~23% to MobA), so global identity cannot distinguish them — but this *local* motif lies exactly in the specificity-determining region. Neumann et al. demonstrated that **"exchange of the complete N-terminal domain of each protein resulted in the total inversion of nucleotide specificity activity, showing that the N-terminal domain determines nucleotide recognition and binding"** (PMID: 21081498), and that as few as five residues within it govern guanine-vs-cytosine recognition. PP_4230's MocA-like loop therefore provides independent, residue-level support for **CTP specificity**.

Mechanistic Model / Interpretation

Molybdenum cofactor biosynthesis in bacteria proceeds through four conserved stages: (1) conversion of 5'-GTP to cyclic pyranopterin monophosphate (cPMP); (2) sulfur insertion to form molybdopterin (MPT); (3) molybdenum insertion to form Moco (Mo-MPT); and (4) optional nucleotide modification, in which either GMP or CMP is attached to the MPT terminal phosphate to make the dinucleotide cofactor variants MGD or MCD ([PMID: 32239579](#); [PMID: 23201473](#)). **PP_4230 acts at step 4, on the CTP/MCD branch**, dedicated to a xanthine-oxidase-family client.

Moco biosynthesis (cytoplasm)

5'-GTP → cPMP → MPT → Mo-MPT (Moco)



↓
Cofactor sulfuration (L-Cys desulfurase / IscS-type)

↓
XdhC/PaoD chaperone (PP_4231, K07402)
binds sulfurated MCD, protects it

↓
Insertion into apo-Mo-subunit PP_4234

↓
Mature XO-family aldehyde oxidase / xanthine dehydrogenase
(PP_4234 Mo-subunit + PP_4233 [2Fe-2S] + PP_4232 multiheme cyt c)

↓ electrons
substrate (aldehyde / purine) oxidation

Where in the cell: PP_4230, like the rest of the cytoplasmic Moco machinery, operates in the **cytoplasm**, producing MCD that is handed to the cytoplasmic XdhC-type chaperone. The *client* enzyme it serves, however, has a divergent fate. PP_4232 carries an N-terminal signal peptide, and the closest structural analogue PaoABC is a *periplasmic* aldehyde oxidoreductase ([PMID: 27622978](#)). This suggests the mature PP_4232–PP_4234 enzyme, once loaded with cofactor in

the cytoplasm, is exported (likely Tat-dependently, as is typical for cofactor-loaded periplasmic molybdoenzymes) to perform aldehyde/purine oxidation in the periplasm. PP_4230 itself, though, is strictly a cytoplasmic cofactor-maturation enzyme.

Pathway logic: The existence of dedicated, per-client MocA paralogs (PP_4230, PP_2483) alongside a single housekeeping MobA (PP_3457) reflects the acceptor-specificity determinant encoded in the C-terminal domain ([PMID: 21081498](#)). Encoding a bespoke cytidylyltransferase within each molybdoenzyme operon ensures that the correct MCD cofactor is delivered to the correct apo-enzyme, and co-regulates cofactor supply with client-enzyme expression. This is a recurring theme in molybdoenzyme maturation: dedicated chaperones and cofactor-modifying enzymes are frequently co-encoded and co-regulated with their client (e.g., tungsten-specific maturation pathways with dedicated MoeA/MobB platforms, [PMID: 42031172](#)).

Evidence Base

PMID	Title (abbrev.)	How it supports the annotation
19542235	<i>MocA is a specific cytidyltransferase in MCD biosynthesis in E. coli</i>	Defines the reaction: MocA + Mo-MPT + CTP + MgCl ₂ → MCD; establishes CTP specificity and that MocA is essential for MCD-containing XO enzymes (YagTSR, XdhABC, XdhD). Primary biochemical anchor for Findings 1–2.
21081498	<i>Amino acid residues determining guanine vs. cytosine specificity</i>	Establishes MocA vs. MobA as CTP- vs. GTP-specific paralogs (22% identity); shows N-terminal domain determines nucleotide specificity and C-terminal domain determines acceptor binding . Anchors Findings 1, 4, 5.
27622978	<i>E. coli periplasmic aldehyde oxidoreductase PaoABC</i>	Structurally characterizes PaoABC as an MCD-containing, periplasmic, heterotrimeric XO-family enzyme — the acceptor-enzyme archetype for PP_4231–PP_4234. Supports Findings 2–3.
16597619	<i>R. capsulatus XdhC Moco binding/insertion</i>	XdhC binds Moco stoichiometrically and inserts it into apo-XDH — defines the function of PP_4231. Supports Finding 3.
10217763	<i>Role of XdhC in Moco insertion</i>	<i>xdhC</i> mutant XDH lacks MPT (but retains FAD/FeS) — XdhC is a specific MPT insertase/chaperone. Supports Finding 3.
17649978	<i>Cysteine desulfurase sulfurates Moco on XdhC</i>	Moco is sulfurated while bound to XdhC before insertion into XDH ; defines the cofactor-sulfuration step downstream of MocA. Supports Finding 3 / mechanistic model.
41933999	<i>Constitutive xanthine oxidase in a P. putida cell factory</i>	In <i>P. putida</i> KT2440, the XdhC-type factor yields ~2.6× more molybdenum and higher activity — experimental validation, in the target organism, that XdhC-type maturation governs XO cofactor loading. Supports Finding 3.
32239579	<i>Moco biosynthesis in E. coli (review)</i>	Defines the 4-step Moco pathway and the CMP/GMP dinucleotide-modification step. Frames the mechanistic model.
23201473		

PMID	Title (abbrev.)	How it supports the annotation
	<i>Molybdenum enzymes and Moco biosynthesis (review)</i>	Comprehensive review of Moco biosynthesis and dinucleotide-variant formation; >50 bacterial molybdoenzymes. Frames the mechanistic model.
42031172	<i>Tungsten-specific maturation pathway</i>	Illustrates the general principle of dedicated, metal/client-specific cofactor-maturation platforms co-encoded with their client enzyme. Contextual support for Finding 4.

Consistency of the evidence: All ten primary/review sources are mutually consistent and point to the same annotation. The strongest experimental anchors (PMIDs 19542235, 21081498) directly characterize the MocA prototype; the XdhC papers (16597619, 10217763, 17649978) define the downstream chaperone step; and

P 41933999

supplies organism-matched (*P. putida* KT2440) experimental validation of the maturation logic. No source contradicts the assignment.

Limitations and Knowledge Gaps

- 1. No PP_4230-specific experimental study exists.** The entire annotation is inferential. There is no published enzyme assay, knockout, structure, or complementation experiment for PP_4230 itself. The molecular-function call rests on orthology (KEGG K07141), domain architecture, an N-terminal specificity motif, and homology to the *E. coli* MocA prototype.
- 2. CTP specificity is inferred, not measured.** While the N-terminal "AAGSSVR" motif matches MocA and not MobA, and while KEGG assigns K07141, the actual nucleotide preference (CTP vs. GTP) and kinetic parameters (Km, kcat) of PP_4230 have not been determined experimentally. It remains formally possible — though unlikely given the motif — that the *P. putida* enzyme has shifted specificity.
- 3. The specific client (PP_4234) is assigned by genomic context and STRING, not by direct interaction data.** The dedicated-partner model is well-supported by the operon structure and the C-terminal specificity principle, but a direct PP_4230–PP_4234 (or PP_4230–PP_4231) protein-protein interaction or cofactor-transfer assay has not been reported.

4. **The physiological substrate of the client enzyme PP_4234 is unknown.** It is annotated broadly as "aldehyde oxidase / xanthine dehydrogenase." Whether it oxidizes purines (xanthine/hypoxanthine), aromatic aldehydes, or another substrate — and hence the metabolic pathway PP_4230 ultimately serves — is undetermined for KT2440.
 5. **Redundancy and essentiality are untested.** With two MocA paralogs (PP_4230, PP_2483) plus MobA (PP_3457), the degree of functional overlap and whether PP_4230 is conditionally essential (e.g., only when the neighboring molybdoenzyme is required) is unknown.
 6. **Localization of the mature client is inferred.** The signal peptide on PP_4232 and homology to periplasmic PaoABC suggest periplasmic function of the mature enzyme, but the export mechanism (Tat vs. Sec) and final localization have not been experimentally confirmed in KT2440.
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Proposed Follow-up Experiments / Actions

1. **Direct enzymatic characterization.** Heterologously express and purify PP_4230; assay nucleotidyltransferase activity with Mo-MPT and CTP vs. GTP in the presence of Mg^{2+} ; determine K_m/k_{cat} and confirm CTP specificity. Compare directly against PP_2483 and PP_3457 (MobA).
2. **Genetic dissection.** Construct in-frame deletions of PP_4230 (and separately PP_2483) in *P. putida* KT2440; assay xanthine dehydrogenase / aldehyde oxidase activity of the PP_4232–PP_4234 enzyme and its molybdenum/MCD content. Test cross-complementation to probe redundancy between the two MocA paralogs.
3. **Cofactor analysis.** Purify the mature PP_4232–PP_4234 enzyme and confirm by LC-MS that it carries the **MCD** (cytosine dinucleotide) form of Moco, as predicted, rather than MGD or Mo-MPT.
4. **Protein-protein interaction mapping.** Use pull-downs, bacterial two-hybrid, or crosslinking to test the predicted PP_4230 ↔ PP_4231 (XdhC) ↔ PP_4234 hand-off, and to verify that the C-terminal domain of PP_4230 confers client specificity (domain-swap with PP_2483).

5. **Substrate identification for PP_4234.** Screen candidate substrates (xanthine, hypoxanthine, aromatic and aliphatic aldehydes) to define the physiological reaction and the metabolic pathway PP_4230 supports.
 6. **Structural validation.** Solve or refine the PP_4230 structure (experimentally, or via the pLDDT-91.9 AlphaFold model) and dock CTP into the N-terminal Rossmann-like domain to confirm the cytosine-recognition residues within the "AAGSSVR" loop.
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Conclusion

PP_4230 (Q88F68) is best annotated as a **dedicated MocA-type CTP:molybdopterin cytidyltransferase** (EC 2.7.7.76, K07141) that catalyzes the Mg^{2+} -dependent, CTP-specific attachment of CMP to Mo-molybdopterin to form the **MCD** variant of the molybdenum cofactor. It functions in the **cytoplasm** as the terminal nucleotide-modification step of Moco biosynthesis, acting as a **client-dedicated paralog** (distinct from housekeeping MobA/PP_3457) that supplies MCD — through the co-encoded XdhC/PaoD chaperone PP_4231 — for the maturation of the neighboring **xanthine-oxidase-family aldehyde oxidase / xanthine dehydrogenase** built around the Mo-subunit PP_4234. The molecular-function assignment is high-confidence; the specific-client assignment is strong but inferential, and both would benefit from the direct experiments proposed above.

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